

Cyber World Modeling

The words all three tracks use, and the places they use them differently

Three tracks sit under this heading and they share a vocabulary nobody negotiated. FOE-Dreamer built the factored model. Learn Structure asks where in a game the strategic part actually lives. AcceleratePSRO asks what a wrong model does to a game solver. Each page was written at its own time, and each took its words from a different literature, so the same word arrives three times carrying three histories.

This file holds the cluster's shared entries. A term belongs here when more than one track uses it. The project file then carries only what that project fixes about it, and points here for the rest. A term whose sense differs across two tracks stays in both files, and each entry says how the other track uses it. That difference is a finding. I am not tidying it away.

Contents

1	The frame	2
1.1	Two challenges, and what they make of the vocabulary	2
1.2	Who does what	2
2	Active	3
2.1	The representation words	3
2.2	The failure words	4
2.3	The allocation words	7
2.4	The budget words	8
3	Working	9
4	Where the tracks disagree	11
5	The four questions, and the terms they will need	12
6	What each project file carries instead	13
7	Sources	13

1 The frame

1.1 Two challenges, and what they make of the vocabulary

The cluster page states two challenges and states them before anything else, which is the right order, because every term below is an answer to one of them.

“Everything a world model has to explain between one step and the next comes from more than one processes that behave nothing alike. The network’s own dynamics are close to stationary and roughly causal: reachability, topology, load.”

website, Cyber World Modeling, Challenge I

“Fold both into a single latent and they contaminate each other: an error in the opponent model bleeds into the estimate of the network, and an ordinary change in the network gets read as the adversary doing something deliberate.”

website, Cyber World Modeling, Challenge I

“Operationally, the stream a model learns from is degraded, intermittent, and partial: sensors are noisy or poisoned, the adversary spends genuine effort staying unobserved, and the gaps are not random but chosen by someone who profits from the defender’s blind spots.”

website, Cyber World Modeling, Challenge II

“The representation has to be learnable from sparse, actively-hidden observation and cheap enough to query inside the control loop”

website, Cyber World Modeling, Challenge II

“You cannot model, or defend, everywhere at once.”

website, Cyber World Modeling, Challenge II, closing line

The first challenge is why the word *factoring* carries weight in this cluster and why *contamination* is the failure it names. The second is why *operational* is stipulated narrowly rather than used as an adjective of praise, and why every acceleration term in the third track is a term about a budget. The closing line is the one sentence the whole cluster turns on, and it is the reason Learn Structure exists at all.

1.2 Who does what

One sentence of mine already divides the labour, and it is the scoping line for this file.

“FOE-Dreamer supplies the factored world model, Learn Structure says where in the game strategic reasoning is worth spending, and Next carries the open questions this track sits inside.”

website, AcceleratePSRO, Related

Read as a vocabulary statement rather than as a table of contents: FOE-Dreamer owns the representation words, Learn Structure owns the allocation words, and AcceleratePSRO owns the budget words. Nearly every disagreement recorded in Section 4 is a word that crossed one of those two boundaries and kept its old sense on arrival.

2 Active

Terms I have to define on the spot, in a talk or in a response, without looking anything up. All four fields.

2.1 The representation words

world model

No page of mine gives this a one-sentence definition, which is itself worth recording, since it is the term the cluster is named for.

Definition: A learned model of the transition dynamics and the reward, queried by rolling it forward instead of acting in the environment.

Distinguishes it from: A simulator, which is written by a person and is correct by construction where it is correct at all. A world model is fitted, so it is wrong somewhere, and where it is wrong is a property of the data it saw. It also differs from an opponent model, which predicts one agent rather than the world the agent acts in.

Operational test: Ask what produced it. If a person specified the transition rules, it is a simulator. If a loss was descended on interaction data, it is a world model, and the next question is where it was fitted.

In my project: FOE-Dreamer builds one and reports it. Learn Structure uses one as the instrument that recovers planted structure. AcceleratePSRO uses one as a substitute for environment interaction inside a solver. The three uses put three different demands on the same object: be accurate, be readable, be cheap.

factoring, factorization

“Instead of one monolithic state it separates the environment’s dynamics from the adversary’s, so the opponent model stays separable and inspectable and an error there is contained rather than smeared across the whole representation.”

website, Cyber World Modeling, Year-1

“The factorization is learned rather than hand-specified.”

website, Cyber World Modeling, Year-1

Definition: The split of a latent state into components by the source of the variation each carries, so that an error in one component is contained inside it.

Distinguishes it from: A factored MDP, where the factoring is a property of the game as written and the parts are state variables. Here the parts are sources of variation, the split is learned, and the argument for it is error containment rather than compactness.

Operational test: Perturb one component and ask what else moves. If everything moves, the split is nominal.

In my project: FOE-Dreamer factors by driver: defender-driven, environment-driven, opponent-inferred. Learn Structure factors per device and keeps two attention graphs so environmental

correlation is not read as strategic dependence. AcceleratePSRO inherits whichever factoring the model it accelerates happens to carry.

latent

Used as a count noun throughout the cluster, which is the usage that matters.

“a Dreamer-style latent world model that keeps the opponent in a latent of its own”

website, Cyber World Modeling, Year-1

Definition: A learned state variable that is not observed, carried forward by a recurrent model, and read by the decoders.

Distinguishes it from: A belief, which is a distribution over a state space that already exists and is named. A latent has no referent fixed in advance, and what it holds is settled by the loss and by whatever separation the architecture enforces.

Operational test: Ask whether the quantity has a name outside the model. A compromise flag has one. A latent does not, and calling it the opponent is a claim about identifiability rather than a description.

In my project: FOE-Dreamer carries the controllable latent, the exogenous latent, and the opponent embedding. Learn Structure carries the same three per device. The cluster word for the third is *opponent latent*.

opponent latent, opponent model

“a compact representation of who the agent is up against”

website, FOE-Dreamer, on the left side of the architecture

Definition: The component of the representation that carries the adversary’s contribution, inferred from defender-side observation and read by whatever consumes it.

Distinguishes it from: Opponent modeling in the game-theoretic sense, which returns a strategy or a type for a player, and threat modeling, which returns a catalogue of what an adversary could do. This one returns a vector and its warrant is that the decoders improve when it is present.

Operational test: Remove the component and retrain. If the reconstruction and the reward prediction do not degrade, nothing about the opponent was in it.

In my project: FOE-Dreamer names it and ablates it. Learn Structure conditions its strategic attention on it. AcceleratePSRO does not carry one, because in a solver the opponent is a member of the strategy population rather than a latent.

2.2 The failure words

drift

The cluster carries two senses of this word, both of them mine, in two neighbouring paragraphs of one page.

“model exploitation, where a policy drifts into states the model never saw and profits from the

error there”

website, FOE-Dreamer, ongoing work

“detecting when the opponent latent has drifted”

website, Next

Definition: Movement of the opponent latent away from the region the model was fitted on.

Distinguishes it from: Distributional shift of the policy, which is the model-based reinforcement learning sense in the first quotation. There the learner does the moving. Here the adversary does.

Operational test: Ask who moved. If the training loop moved, the sense is the borrowed one. If the other side moved, the sense is mine.

In my project: FOE-Dreamer plans a drift detector on the opponent latent and a fallback detector on the joint latent. Learn Structure has the same phenomenon under a different name, the manipulation schedule, because there the experimenter moves the opponent deliberately and on a known schedule.

objective mismatch

“objective mismatch, where the model minimises prediction error across the data distribution while the policy needs accuracy only where it will act”

website, FOE-Dreamer, ongoing work

Definition: The gap between the loss the model descends and the accuracy the policy needs, since the first is averaged over the data and the second is local to where the policy acts.

Distinguishes it from: Underfitting, which is failure on the same distribution the loss is averaged over. Objective mismatch is compatible with a low training loss and is caused by it.

Operational test: Compare the model’s error on the data distribution against its error on the states the policy visits. If they differ and the second is worse, the mismatch is live.

In my project: FOE-Dreamer names it as one of the two borrowed failures. AcceleratePSRO inherits it in the sharpest possible form, because the states a best response visits are the states the previous iteration did not.

model exploitation

“a policy drifts into states the model never saw and profits from the error there”

website, FOE-Dreamer, ongoing work

Definition: A policy improving its modelled return by moving into states where the model is wrong.

Distinguishes it from: Objective mismatch, which is a property of the loss. Model exploitation is a property of the optimizer’s answer to it. The two travel together and are not the same failure.

Operational test: Score the policy in the model and in the environment. A gap that widens with the number of policy improvement steps is exploitation.

In my project: Written down on the FOE-Dreamer page as a borrowed failure, with the twist that makes it a cluster term rather than an imported one.

the twist: drift as an opponent’s objective

“In both the drift is an artefact of the training loop. Here it is an opponent’s objective, so uniform predictive loss is selected against.”

website, FOE-Dreamer, ongoing work

“the adversary is the process that looks for the region where the model is confidently wrong and operates there”

website, Next

Definition: The claim that in an adversarial setting the two borrowed failures stop being accidents of optimization, because a party outside the training loop is searching for the same region the optimizer would have stumbled into.

Distinguishes it from: The single-agent versions, where the drift is produced by the learner and can be bounded by the learner’s own update rule. No update rule bounds a second party.

Operational test: Ask whether the process producing the shift has an incentive. If it does, no amount of care inside the training loop closes the gap.

In my project: This is the sentence that connects the two open questions on the Next page, and the reason the third track and the first are studying one thing.

fails honestly, fails silently

“preferring a model that fails honestly (wide and uncertain) over one that fails silently (sharp and wrong)”

website, Next

Definition: Two failure shapes for a learned model. Honest failure is wide predictive uncertainty in the region the model is wrong about. Silent failure is a sharp prediction that is wrong.

Distinguishes it from: Calibration in the ordinary sense, which is a property averaged over a whole test set. This distinction is local, and only the region an adversary would choose is being asked about.

Operational test: Find a region the model is wrong about, then read its predictive spread there. Wide is honest, sharp is silent.

In my project: The FOE-Dreamer page states the same idea as a question, whether a model with wide uncertainty is preferable to a confident one that is wrong. The Next page states it as the coinage, and the coinage is better, because it says what the two failures look like rather than which is preferred.

identifiability

“the adversary latent may not be identifiable from defender-side observation at all”

website, FOE-Dreamer, ongoing work

“The fallback is drift detection on the joint latent, which surrenders attribution and keeps the alarm.”

website, FOE-Dreamer, ongoing work

Definition: Whether the adversary’s contribution is recoverable from the data one side actually holds, separately from everything else in the latent.

Distinguishes it from: Accuracy. A model can predict well and still hold no separable adversary component, since fitting the joint law does not constrain its factorization.

Operational test: Vary the opponent while holding the environment fixed. If the component that is supposed to carry the opponent does not move, it is not carrying it.

In my project: FOE-Dreamer states the assumption and states the fallback in the same breath, as a trade with attribution on one side and the alarm on the other. Learn Structure declares three limits of the same shape, one of which is that dependence no configuration of the schedule exercises is invisible in principle from one side’s data.

2.3 The allocation words

strategic dependence

This is the term with two of my own phrasings, and both are quoted in Section 4. The cluster sense is the one on the Learn Structure page.

“Strategic dependence is how much a player’s best action, in a given region of states and actions, turns on what the other player does.”

website, Learn Structure, the intuition

Definition: The degree to which one player’s best choice depends on another player’s behaviour, in a given region of states and actions.

Distinguishes it from: The dependency graph of a factored MDP, the relations of an object-oriented state, the causal graph, and empirical payoff dependence, all four of which are properties of the game as written. This one is a property of how the game is played, and it moves while the game does not.

Operational test: Fix a game completely and change only one player’s behaviour. If the other player’s best action changes, dependence is present at that region and none of the four neighbours can express it.

In my project: Learn Structure defines it, classifies it by five mechanisms, and grades each mechanism on an ordered scale. The cluster page gives it a second phrasing, as value forfeited. FOE-Dreamer does not use the term, and its opponent latent is the representation-side answer to the same question.

non-stationary

“it adapts to the defender, it hides, and it shifts precisely when the defender starts to model it well”

website, Cyber World Modeling, Challenge I, on the adversary’s contribution

Definition: A transition law that changes over the interaction because a party outside the model is changing its own behaviour.

Distinguishes it from: Endogenous environment change, where the world moves on its own for

reasons of its own, a device wearing out or a service churning. Both violate stationarity and only one of them is answering the defender.

Operational test: Ask whether the change tracks the defender’s own improvement. The clause about shifting when the defender starts to model it well is the discriminating one.

In my project: Challenge I of the cluster page. Learn Structure buys stationarity back by fixing the other two parties within a batch, and states the price openly: every learned quantity is relative to the manipulated policies in force when the data was collected.

operational

The one term in the cluster whose scope I stipulate rather than assume, and the whole entry is the stipulation.

“Narrowly. The substrate, the services, the exploit paths, the five-second polling and the consequences of defender actions are real. The user and attacker populations and the CVE catalogue are emulated.”

website, FOE-Dreamer, what operational means here

Definition: A claim about a specific list of things being real, given with the complementary list of things that are emulated.

Distinguishes it from: Real-world, production, and deployed, none of which carry a list, and all of which a reader will read as a stronger claim than the one being made.

Operational test: Ask for the two lists. A use of the word that cannot produce them is doing rhetorical work rather than descriptive work.

In my project: Six items real, three emulated, on the FOE-Dreamer page. The same discipline is what Learn Structure applies to its generated instances, where every reported quantity is either set by construction or measured, and a request that is neither is refused with its infeasibility named.

2.4 The budget words

interaction budget

Definition: The number of environment steps a method is allowed before its result is read.

Distinguishes it from: Compute budget, which counts gradient steps and wall clock. The two come apart exactly where a world model is used, since imagined rollouts cost compute and no interaction.

Operational test: Ask what the horizontal axis of the plot counts. If it counts environment steps, the budget is interaction. If it counts hours, it is compute, and a model-based method will look better on one and worse on the other.

In my project: AcceleratePSRO measures regret against interaction budget. FOE-Dreamer reports both, a three-day budget on one GPU and matched compute against the baselines, because the deployment argument is about interaction and the fairness argument is about compute.

imagination, imagined rollout

“imagined rollouts can stand in for expensive environment interaction when computing a best response inside PSRO”

website, AcceleratePSRO, the intuition

Definition: A trajectory generated by rolling the learned model forward rather than by acting, used wherever an environment trajectory would have been.

Distinguishes it from: Simulation, which rolls a hand-written model forward. The word imagination is reserved for a learned model, and the reason the distinction is kept is that only one of the two can be wrong in a way nobody specified.

Operational test: Ask whether the generator of the trajectory was fitted. If it was, the trajectory carries the model’s error into whatever consumes it.

In my project: AcceleratePSRO asks when imagination pays and names the crossover point below which it stops. FOE-Dreamer trains its policy on imagined rollouts as a matter of course, which is what makes the first track’s model the third track’s instrument.

regret

Definition: The payoff a player gives up by playing its chosen strategy rather than the best available response, read against a stated set of alternatives.

Distinguishes it from: Episode loss, which is FOE-Dreamer’s metric and is an absolute quantity against no alternative. Regret needs a comparison class and is meaningless without one.

Operational test: Name the set the maximum is taken over. If nobody can name it, the number is not a regret.

In my project: AcceleratePSRO uses the empirical game-theoretic sense, regret of a profile against a strategy set. Learn Structure names regret per ordered pair of players as one of the two holes nothing in its measure set fills. Those are different quantities and both are called regret.

3 Working

Recognized precisely, lookup acceptable. Definition and discrimination only.

Daedalus

“an eight-host OpenStack network across three subnets, provisioned with real services, with CVE-bound exploits and GHOSTS-generated background user traffic”

website, FOE-Dreamer, results

Distinguishes it from: A simulator such as CyGym or CybORG. Daedalus is substrate, so an action’s consequence is produced by the machines rather than by a transition rule.

CyGym

“a simulation-based game-theoretic analysis framework for cybersecurity, which supplies the empirical game the loop is solved over”

website, AcceleratePSRO, environment

Distinguishes it from: Daedalus, which is emulation. The cluster runs on both deliberately, and the two are not interchangeable: Daedalus can carry a deployment claim and cannot carry a game solved to equilibrium, and CyGym is the reverse.

GHOSTS

Definition: The traffic generation framework whose scripted personas produce the benign user activity in Daedalus.

Distinguishes it from: Injected noise. The personas act, so the resulting activity is correlated with the state rather than added to it.

edge hardware

“the model also has to run where the decision is taken, on edge hardware with a fraction of the compute a latent world model would prefer”

website, Cyber World Modeling, Challenge II

Distinguishes it from: A training budget. This is an inference constraint inside a control loop, and it is one of the two reasons the cluster treats factoring as a requirement rather than as an elegance.

living off the land

Definition: An adversary operating with the tools the host already provides, so that its actions are indistinguishable from administration.

Distinguishes it from: Stealth by evasion, which hides an artefact. This hides by producing no artefact that differs from a legitimate one, and it is the case in which the adversary latent may not be identifiable at all.

planted structure, known ground truth

“planted structure is recovered against known ground truth”

website, FOE-Dreamer, success criteria

Distinguishes it from: Held-out prediction accuracy. Recovery is checked against a value that was written by the generator rather than measured on the data, which is only available in a generated instance and is why Learn Structure builds one.

drift detector, failure taxonomy

Definition: The two planned outputs of the FOE-Dreamer ongoing work: an alarm that fires on movement of the opponent latent, and a classification of the ways a factored world model goes wrong.

Distinguishes it from: Anomaly detection on the observation stream, which fires on the world changing. These fire on the model’s own representation changing.

Rainbow, IQN

Definition: The two model-free baselines FOE-Dreamer is measured against under matched

compute.

Distinguishes it from: Model-free PSRO, which is AcceleratePSRO’s baseline and is not a policy learner but a solver. The two tracks compare against different things because they claim different things.

4 Where the tracks disagree

Five places where one word does two jobs across my own files. Each sits here rather than in an errata, because a term whose two uses disagree is the entry most worth having.

Strategic dependence, defined twice

The Learn Structure page: strategic dependence is *how much a player’s best action, in a given region of states and actions, turns on what the other player does*.

The cluster page, one level up, in the paragraph that introduces the same project: *how much value a defender forfeits by ignoring the opponent, as a function of the game’s structure*.

The first is sensitivity of a best action. The second is a loss in payoff units. They point at the same phenomenon and they are not the same quantity, and the difference matters exactly where the taxonomy is thinnest, since one of the two recorded holes in the measure set is a pairwise deviation gain in payoff units. Until that hole is filled, the cluster page is naming a quantity the project page does not yet define.

Adversary, opponent, attacker, poacher

Four words for the other side, and no page says whether the difference is intended. The cluster page uses adversary for the source of non-stationary change and opponent for the model of it, in the same paragraph. FOE-Dreamer says scripted attacker profiles for the two behaviours in Daedalus and opponent latent for what the model infers. Learn Structure says poacher in its worked counterexample and attacker in its network domain.

The reading that fits every occurrence is that adversary names the party, opponent names the model of the party, and attacker and poacher name the role in a specific domain. It fits, but it is not stated anywhere, and a reader arriving from any one page will read the other two through their own use.

Structure, in a cluster that names one project after it

Learn Structure’s own glossary carries seven senses of the word and says so: *The word structure carries a different precise meaning in each literature we draw on*. Factored, abstraction, bisimulation, object, causal, empirical game, and strategic dependence.

FOE-Dreamer uses the word in a sense none of the seven covers, the split of a latent by the source of its variation. AcceleratePSRO uses it barely at all, and where it does the referent is the empirical game’s payoff structure.

So the cluster’s second project is named after a word that the cluster’s first project uses differently, and the shared page never reconciles them. The reconciliation, if one is wanted, is that FOE-Dreamer’s sense is the learned counterpart of the object and causal senses, applied to a latent rather than to observed variables.

The environment, in three incompatible senses

FOE-Dreamer’s environment is Daedalus, a substrate on which things happen. Learn Structure’s environment is a generated instance whose every dependence is recorded with the grounds on which its value is known. AcceleratePSRO’s environment is CyGym, whose job is to supply an empirical game.

Three tracks, three environments, and the word carries an assumption in each case about what can be claimed from a result obtained in it. The first supports a deployment claim. The second supports a recovery claim. The third supports an equilibrium claim. None supports the others, and a sentence in the cluster’s voice that says the work runs in realistic environments is true of one third of the cluster at a time.

Model error, priced three ways

FOE-Dreamer prices model error as a defence failure: which errors a defender survives, and whether the drift is caught before the region is reached.

Learn Structure prices it as a recovery failure, and splits it in two, since a model can fit the data and not the law, and it can fit the law while mislabelling who drives whom.

AcceleratePSRO prices it as a solver failure: model bias becomes bias in the best-response target, which the game-solving step then amplifies.

One quantity, three consumers, three different questions asked of it. This is the reason the cluster cannot report a single number for how good its world model is, and the reason the first ongoing question on the cluster page is the one it is.

5 The four questions, and the terms they will need

The cluster page closes on four questions. Two are carried into the Next page and two are not, and the two that are not are the ones with no vocabulary yet.

1. Whether a world model is reasonably accurate, and which errors a defence agent in an operational network can accept. The terms are here already: honest against silent failure, drift, the failure taxonomy, the success criteria fixed in advance.
2. Whether a reasonably accurate model can expedite the search for strategic responses rather than only roll the world forward. The terms are in `cwm_accelerate_psro.tex`: acceleration levers, the risk register, the crossover point.
3. How to factor a model so that it generalizes to novel scenarios with zero or few-shot training. No page in the cluster develops this, and the vocabulary for it does not exist yet. What exists is the identifiability entry above, which is the negative half of the same question.
4. How to compose a suitable model for a target operational environment, given a set of components. Also undeveloped, and the nearest thing to a vocabulary is Learn Structure’s per-type record, since a component set that can be composed is a component set whose dependences are declared.

Questions three and four appear once each, on the cluster page, and nowhere else. They are the cluster’s open ground.

6 What each project file carries instead

File	Carries
<code>cwm_foe_dreamer.tex</code>	The architecture symbols, the three latents and their separation argument, the testbed, the evaluation, and the eight places two of my own files disagree about the model.
<code>cwm_learn_structure.tex</code>	The seven senses of structure, the five types and their degree scales, the four excuses, the five candidate measures, the generator's record vocabulary, and the two success criteria of the learner.
<code>cwm_accelerate_psro.tex</code>	PSRO and its routines, Dyna-PSRO, CyGym, the acceleration levers, the risk register, the validation protocol, and the crossover point.
<code>dict_population_based_training.tex</code>	The six senses of population, which the cluster uses in two of them at once.
<code>dict_cyber_env.tex</code>	Daedalus, CyGym, and the rest of the environment vocabulary, placed against the field's.

7 Sources

- The five pages of `content/overview/3-year-agenda/cyber-world-modeling/`: the cluster README, `environment.md`, `strategic-structure.md`, `accelerate-psro.md`, and `next.md`. Every quoted sentence above is from one of them, and the locator on each entry says which.
- The Learn Structure notes, for the entries where a project file fixes more than the page does: `Notes/Latex/sections/glossary.tex` for the seven senses of structure, `strategic_dependence.tex` for the separating example, `generator.tex` for construction against measurement and for refusal, and `learn_stage.tex` for the manipulated setting and the two success criteria.
- `_side/_lexicons/cwm_foe_dreamer.tex`, whose closing section names the terms that cross into this file, and `_side/_lexicons/lexicon_method.tex`, for the four fields, the three tiers, and the rule that a term whose sense differs across projects stays in both.
- The Tech Fluency page, for the three fluency levels and the tier definitions.